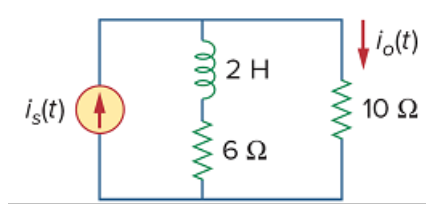


Problem

Find the current $i_o(t)$ in the circuit in Fig. 18.21, given that $i_s(t) = 50 \cos 4t$ A.

Figure 18.21



Step-by-step solution

Step 1 of 3

Refer to Figure 18.21, in the text book.

By current division,

$$\begin{aligned} H(\omega) &= \frac{I_o(\omega)}{I_s(\omega)} \\ &= \frac{6 + j\omega 2}{16 + j\omega 2} \end{aligned}$$

Fourier transform of $A \cos \omega_0 t \leftrightarrow A\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$

Then the Fourier transform of $i_s(t) = 50 \cos 4t$ is $I_s(\omega) = 50\pi [\delta(\omega + 4) + \delta(\omega - 4)]$

Hence, the value of $I_o(\omega)$ is,

$$\begin{aligned} I_o(\omega) &= H(\omega) I_s(\omega) \\ &= \frac{(6 + j\omega 2) 50\pi [\delta(\omega + 4) + \delta(\omega - 4)]}{(16 + j\omega 2)} \end{aligned}$$

Step 2 of 3

Apply inverse Fourier transform,

$$\begin{aligned} i_o(t) &= F^{-1}[I_o(\omega)] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{(6 + j\omega 2) 50\pi [\delta(\omega + 4) + \delta(\omega - 4)]}{(16 + j\omega 2)} e^{j\omega t} d\omega \\ &= \frac{50\pi}{2\pi} \int_{-\infty}^{\infty} [\delta(\omega + 4) + \delta(\omega - 4)] \frac{(6 + j\omega 2)}{(16 + j\omega 2)} e^{j\omega t} d\omega \\ &= 25 \int_{-\infty}^{\infty} [\delta(\omega + 4) + \delta(\omega - 4)] \frac{(6 + j\omega 2)}{(16 + j\omega 2)} e^{j\omega t} d\omega \end{aligned}$$

Step 3 of 3

Apply the shifting property of the impulse function,

$$\int_{-\infty}^{\infty} \delta(\omega - \omega_0) f(\omega) d\omega = f(\omega_0)$$

Here $f(\omega) = \frac{(6 + j\omega 2)}{(16 + j\omega 2)} e^{j\omega t}$

And obtain

$$\begin{aligned} i_0(t) &= 25 \left[\frac{6 - j8}{16 - j8} e^{-j4t} + \frac{6 + j8}{16 + j8} e^{j4t} \right] \\ &= 25 \left[0.559 e^{-j26.6^\circ} e^{-j4t} + 0.559 e^{j26.6^\circ} e^{j4t} \right] \\ &= 13.98 \left(e^{-j(4t+26.6^\circ)} + e^{-j(4t+26.6^\circ)} \right) \\ &= 27.95 \left(\frac{e^{-j(4t+26.6^\circ)} + e^{-j(4t+26.6^\circ)}}{2} \right) \end{aligned}$$

The expression of cosine function is, $\cos \omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$

$$i_0(t) = 27.95 \cos(4t + 26.6^\circ) \text{ A}$$

Therefore, the current $i_0(t)$ is,

$$\boxed{i_0(t) = 27.95 \cos(4t + 26.6^\circ) \text{ A}}.$$